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1 - OVERVIEW

This section applies to Signa Horizon/LX II 1.0T systems installed with the RF/PDU Cabinet and explains the procedures for the replacement of parts for RF amplifiers.

Parts mentioned are listed in Table 1-1:

TABLE 1-1
FRU(1) PART LIST

Manufacturer's Part Number	GEMS Part Number	Item Description
TGA4204A	2170783-2	RF Amplifier with Case
MA77B3	2195572	Fan
LDA15F-24-SN	2195574	24 Volt Power Supply

Special tools needed to replace the amplifier are listed in Table 1-2:

TABLE 1-2
SPECIAL TOOLS NEEDED

Item	Description	Part Number
1	Universal Lift Hoist Kit – Used to remove amp from cabinet.	46-317266G5

2 - LOCKOUT / TAGOUT PROCESS

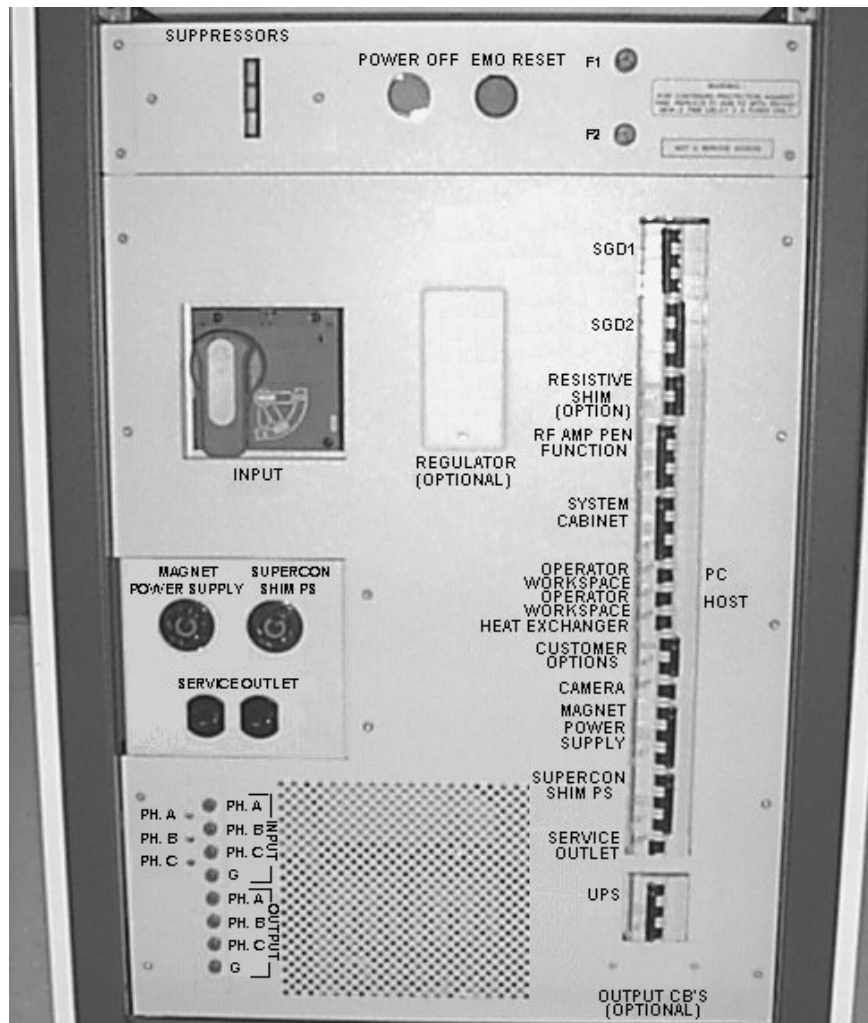


POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE PDU MAIN INPUT CIRCUIT BREAKER BEFORE REMOVING THE RF AMPLIFIER. VERIFY THAT LOCK OUT AND TAG OUT OF THE PDU MAIN INPUT CIRCUIT BREAKER IS PROPERLY PERFORMED.

1. Bring the PC software down.
 - a. Toggle mouse control to the PC monitor using the hardkey on the top, left side of the keyboard housing.
 - b. From the PC monitor use the mouse to select **Start → Shutdown → Shut down**.
2. Bring the Signa software down.
 - a. Toggle mouse control back to the Signa monitor using the hardkey on the top, left side of the keyboard.
 - b. Single-click on the **Toolbelt** icon.
 - c. Single-click on the **System Shutdown** softkey on the Service Desktop Manager.
 - d. Select **OK** to confirm the shutdown.
 - e. Wait for the system to indicate on the monitor that it is safe to power off the computer before proceeding. This usually takes about 90 seconds before this message is seen.
3. Remove the cover from the front of the RF/PDU cabinet to expose the front of the PDU.

2 - LOCKOUT / TAGOUT PROCESS (CONTINUED)

4. Refer to Illustration 2-1 and power off the following circuit breakers:
 - a. **SGD 1**
 - b. **SGD 2**
 - c. **RFI / AMP** (shown as RF Amp Pen Function in Illustration 2-1)
 - d. **SYSTEM SUPPORT MODULE**
 - e. **OPERATORS WORKSPACE (PC)**
 - f. **OPERATORS WORKSPACE (HOST)**
 - g. **GRADIENT WATER CHILLER**



BREAKER LOCATION
ILLUSTRATION 2-1

2 - LOCKOUT / TAGOUT PROCESS (CONTINUED)

5. Rotate the larger main breaker labeled INPUT counter-clockwise to the green, O (OFF) position. See Illustration 2-2.



MAIN INPUT BREAKER IN THE OFF POSITION
ILLUSTRATION 2-2

6. Lock Out / Tag Out the PDU "INPUT" circuit breaker.
 - a. Press on the left-pointing arrow on the end of the handle to expose the location where the lock and tag can be placed.

2 - LOCKOUT / TAGOUT PROCESS (CONTINUED)

- b. Lock and tag out the PDU “INPUT” circuit breaker as shown in Illustration 2-3



PDU MAIN INPUT CIRCUIT BREAKER LOCKED AND TAGGED OUT
ILLUSTRATION 2-3

7. Verify all LEDs are OFF (not illuminated) on front of SSM and RFI. Verify all LEDs are OFF (not illuminated) at the rear of the RF Amplifier Power Module.
8. Using an AC voltmeter (DVM), measure the input power leads on the rear of one of the RF Amplifier Modules from the rear (right side) of the RF/PDU Cabinet.
 - Measure between L1 and L2; verify voltmeter measures 0 VAC.

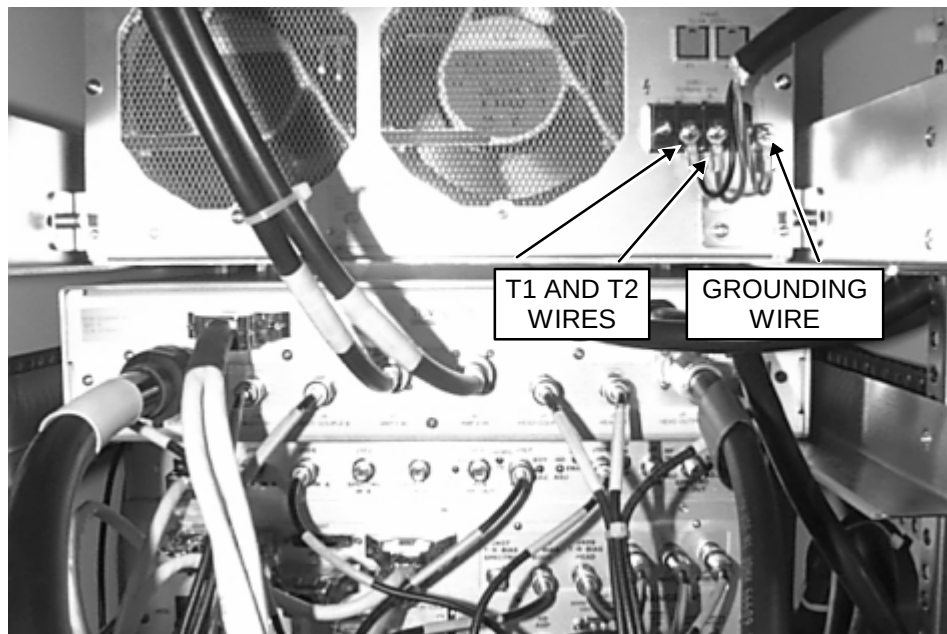
3 - RF AMPLIFIER

3-1 RF Amplifier Power Module Removal

DANGER!!

POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE PDU MAIN INPUT CIRCUIT BREAKER BEFORE REMOVING RF AMPLIFIER. VERIFY THAT LOCK OUT AND TAG OUT OF THE PDU MAIN INPUT CIRCUIT BREAKER IS PROPERLY PERFORMED.

1. Perform Section **2 - LOCKOUT / TAGOUT PROCESS** (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)
2. Remove the J3 and J1 connections to the front of the RF Amplifiers.
3. Disconnect wires from T1 and T2 of 208VAC connectors and the grounding wire on rear of RF AMP. See Illustration 3-1.



T1, T2, AND GROUNDING WIRE CONNECTORS
ILLUSTRATION 3-1

4. Remove the J4 connection to RF Out port.
5. Remove the front cover and open the rear door of the cabinet.
6. Remove the four screws securing the amplifier to the cabinet.

WARNING!

POSSIBLE PERSONAL INJURY. HEAVY OBJECT - THE HOIST KIT WEIGHS 73 LBS/33 KG. ASSISTANCE TO LIFT THIS PRODUCT [MECHANICAL (DOLLIES) OR ADDITIONAL PERSONNEL] IS REQUIRED.

7. Refer to **APPENDIX A - UNIVERSAL LIFT KIT ASSEMBLY FOR 1.0T RF/PDU AND SRF CABINETS** procedure to remove amplifier from cabinet.

3-2 Amplifier Replacement

WARNING!

POSSIBLE PERSONAL INJURY. HEAVY OBJECT - THE HOIST KIT WEIGHS 73 LBS/33 KG. ASSISTANCE TO LIFT THIS PRODUCT [MECHANICAL (DOLLIES) OR ADDITIONAL PERSONNEL] IS REQUIRED.

1. Refer to **APPENDIX A - UNIVERSAL LIFT KIT ASSEMBLY FOR 1.0T RF/PDU AND SRF CABINETS** procedure to install the amplifier into the cabinet.
2. Insert four screws into front of amplifier to secure to cabinet.
3. Reconnect cables to J3 and J1 of the RF Amplifier front.
4. Reconnect J4 RF Out connection to rear of RF Amplifiers.
5. Connect black wire to T1 and red wire to T2 and green/yellow wire to grounding connection on rear of RF Amplifiers.
6. Put the CB1 breaker in the inside, rear of the SRF cabinet in the ON position.
7. Close the SRF rear cabinet door.
8. Remove Lock Out / Tag Out hardware from the PDU "INPUT" circuit breaker and rotate the breaker fully clockwise into the red, 1 (ON) position to restore power to the system.
9. Push the red **EMO RESET** button on the top, front of the PDU once to reset the emergency stop circuit.
10. Place the following circuit breakers on the front of the Power Distribution Unit (PDU) into the right (ON) position:
 - a. **SGD 1**
 - b. **SGD 2**
 - c. **RFI / AMP** (shown as RF Amp Pen Function in Illustration 2-1)
 - d. **SYSTEM SUPPORT MODULE**

e. **OPERATORS WORKSPACE (PC)**

f. **OPERATORS WORKSPACE (HOST)**

g. **GRADIENT WATER CHILLER**

11. Confirm that LEDs are illuminated on the front of the RFI and SSM and confirm that the fans in the RF amplifiers are rotating.
12. Power the Host and PC computers on and then boot the SIGNA and PC software as required.
13. If an amplifier was replaced then perform the RFI calibration procedure described in **1.0T SRF Max Power RF Out Setup and Calibration** (RC3SCA3.DOC) on the 8.X Service Methods CDROM.

4 - AMPLIFIER CAPACITOR BANK SAFETY.

4-1 Tools Required

- Digital Multimeter
- Phillips Screwdriver
- Pliers

4-2 Verifying the Capacitor Bank is Discharged

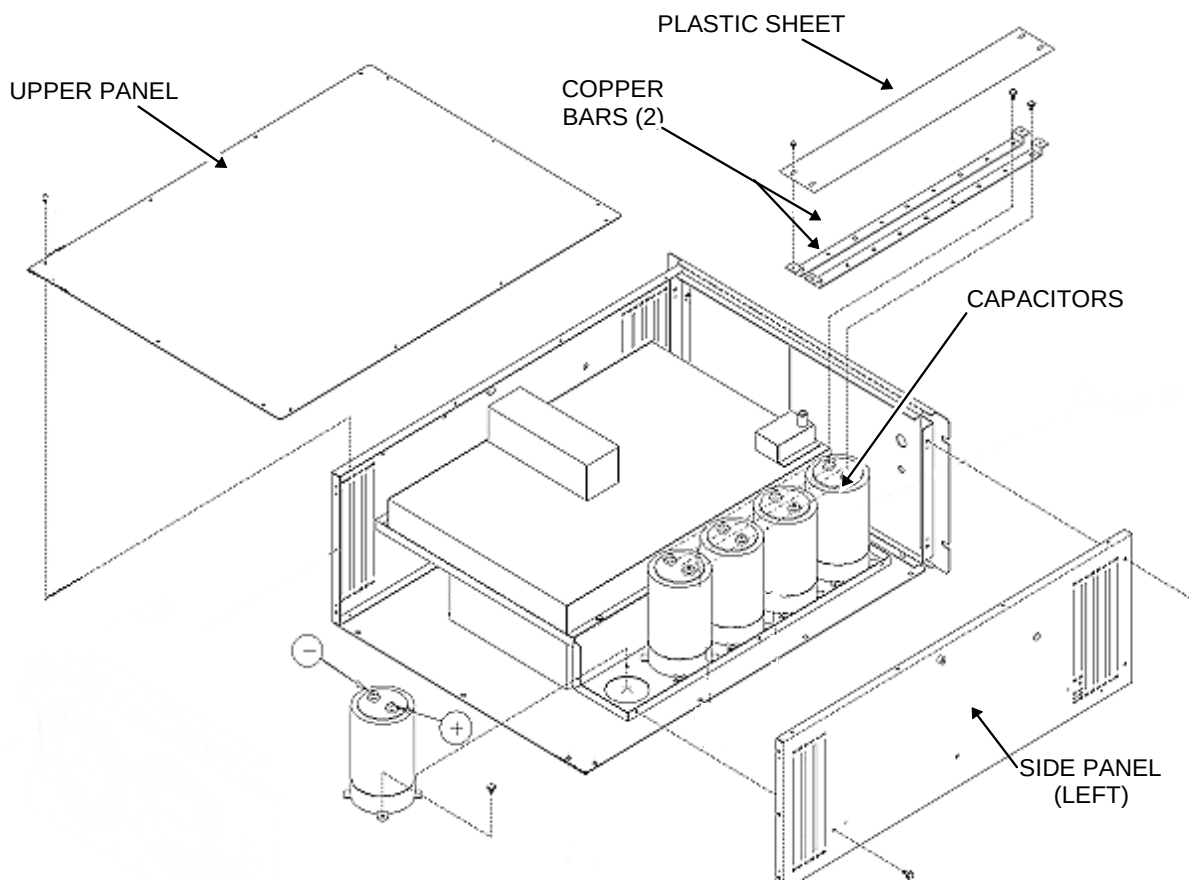


SHOCK HAZARD! AVOID SERIOUS INJURY BY ELECTROCUTION. THE CAPACITOR BANK IN THE AMPLIFIER MAY RETAIN CHARGE LONG AFTER THE INCOMING 208V~ LINES AT T1 AND T2 ON THE REAR OF THE AMPLIFIER HAVE BEEN DISCONNECTED. BEFORE REPLACING ANY COMPONENTS IN THE AMPLIFIER, MEASURE THE CAPACITOR VOLTAGE AS DESCRIBED IN THIS SECTION TO VERIFY THAT THE VOLTAGE IS BELOW 10 VDC AT A SAFE LEVEL. DO NOT ASSUME THAT THE BLEEDER RESISTOR HAS DISCHARGED THE CAPACITOR BANK BASED ON THE STATE OF THE INDICATOR LEDS. ALWAYS MEASURE TO VERIFY. DISCHARGE CAPACITOR BANK, IF NECESSARY, TO LESS THAN 10V BEFORE REMOVING OR REPLACING ANY COMPONENTS IN THE RF AMPLIFIER. SEE THIS SECTION FOR STEPS TO DISCHARGE THE CAPACITOR BANK.

1. Remove RF Amplifier from cabinet. Refer to Section **3-1 RF Amplifier Power Module Removal.**

4-2 Verifying the Capacitor Bank is Discharged (Continued)

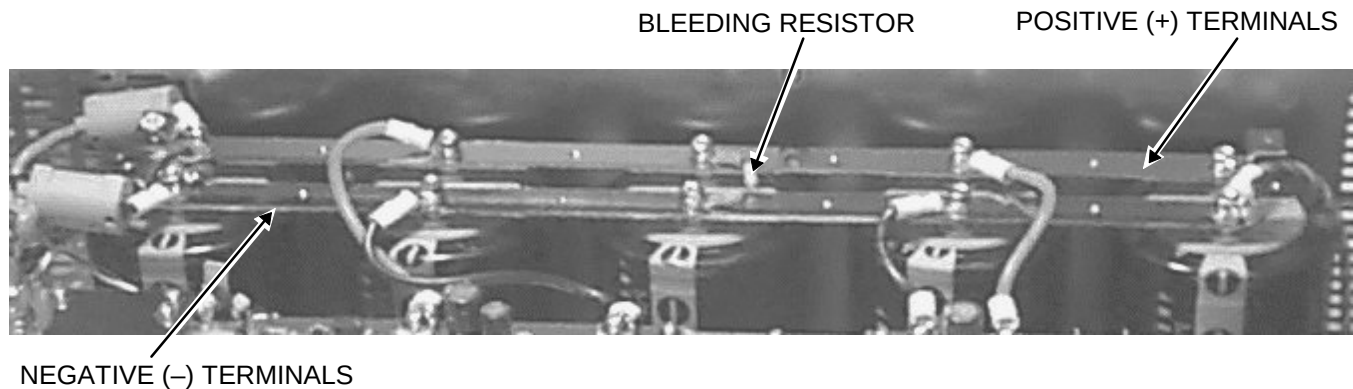
2. Wait 15 minutes from the time power was removed from the amplifier for the internal amplifier capacitor bank to discharge before proceeding to the next step.
3. Remove 14 screws on the upper panel of RF AMP and remove panel. See Illustration 4-1.



DISASSEMBLY OF UPPER PANEL AND CAPACITOR LOCATION
ILLUSTRATION 4-1

4-2 Verifying the Capacitor Bank is Discharged (Continued)

4. There are 5 capacitors joined together at each of the (+) and (–) capacitor terminals with a copper bus bar. See Illustrations 4-1 and 4-2. There is a bleeder resistor mounted across the two bus bars. See Illustration 4-2. There are also secondary bleeder circuits and a voltage indicator LED (LED201) on each PA circuit board. 55 VDC is present between the copper bus bars during normal operation. The measured voltage must be below 10 VDC before working on the equipment. Measure the voltage present between the (+) and (–) copper bars.

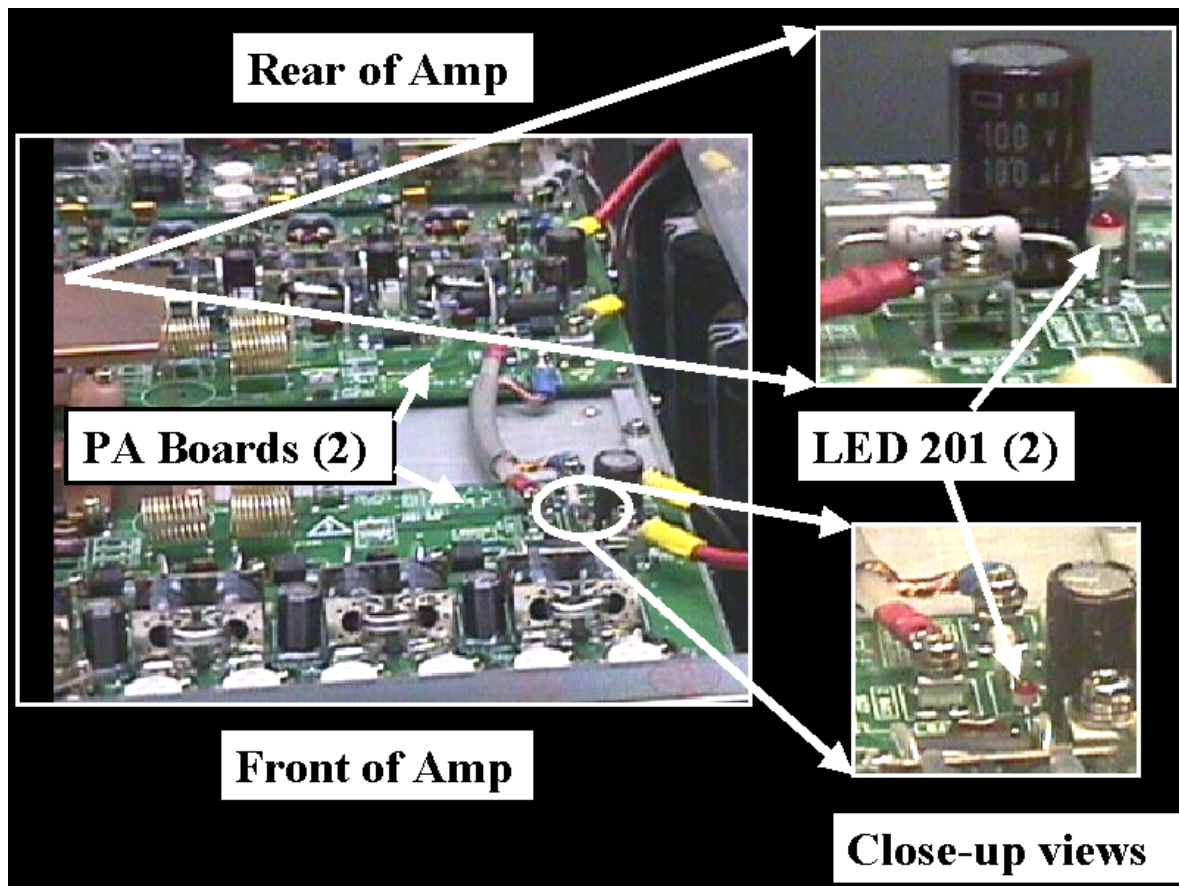


CAPACITOR WIRING
ILLUSTRATION 4-2

- a. The voltage is at a safe level if it is below 10 VDC. If the voltage measures below 10 VDC then go to step 3.

4-2 Verifying the Capacitor Bank is Discharged (Continued)

- b. If the voltage does not measure below 10 VDC then check to see if LED201 on either of the 2 PA circuit boards is illuminated. See Illustration 4-3.



PLACEMENT OF LED201 ON EACH PA CIRCUIT BOARD.
ILLUSTRATION 4-3

- i. If either LED 201's are illuminated then wait for the voltage level to decrease to less than 10 VDC before replacing any components in the amplifier. This usually takes less than 15 minutes from the time the amplifier power is removed but it is possible that it could take longer.
- ii. If neither of the two (2) LED 201's are illuminated and the measured voltage is still greater than 10 VDC then either wait for the capacitor bank to further discharge or obtain a 10,000 ohm, 0.5 watt resistor and connect it between the (+) and (-) copper bars to discharge the capacitor bank down to a level below 10 VDC.
5. When it has been confirmed that the capacitor bank has been discharged to a level below 10 VDC then it is safe to begin work inside the RF amplifier. Proceed to the section of interest.

5 - FAN

5-1 Tools Required

- Phillips Screwdriver (30cm or longer)
- Pliers

5-2 Removal of Fan

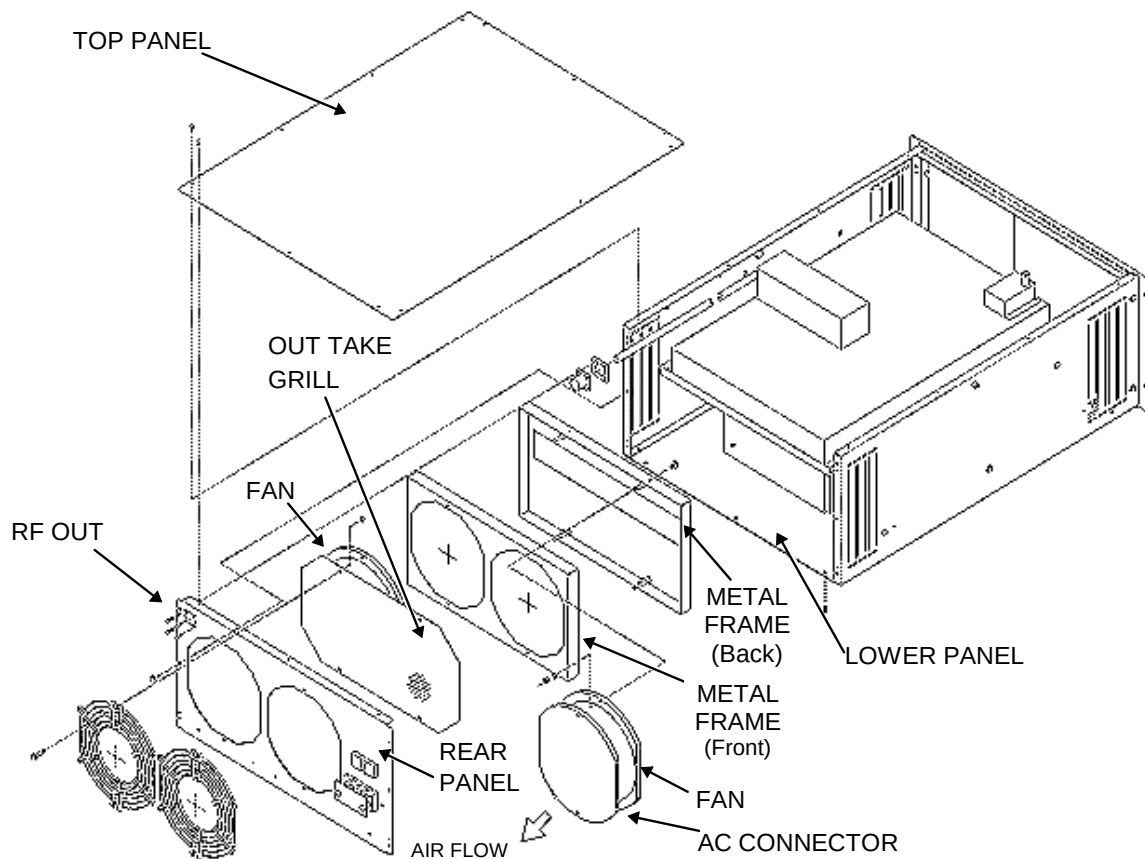


SHOCK HAZARD! AVOID SERIOUS INJURY BY ELECTROCUTION. THE CAPACITOR BANK IN THE AMPLIFIER MAY RETAIN CHARGE LONG AFTER THE INCOMING 208V~ LINES AT T1 AND T2 ON THE REAR OF THE AMPLIFIER HAVE BEEN DISCONNECTED. BEFORE REPLACING ANY COMPONENTS IN THE AMPLIFIER, MEASURE THE CAPACITOR VOLTAGE AS DESCRIBED IN SECTION 4-2 TO VERIFY THAT THE VOLTAGE IS BELOW 10 VDC. DO NOT ASSUME THAT THE BLEEDER RESISTOR HAS DISCHARGED THE CAPACITOR BANK BASED ON THE STATE OF THE INDICATOR LEDS. ALWAYS MEASURE TO VERIFY. DISCHARGE CAPACITOR BANK, IF NECESSARY, TO LESS THAN 10V BEFORE REMOVING OR REPLACING ANY COMPONENTS IN THE RF AMPLIFIER. SEE SECTION 4-2 FOR STEPS TO DISCHARGE THE CAPACITOR BANK.

1. Confirm that the 14 screws on the upper panel of the RF amplifier have been removed and that the capacitor bank has been discharged to a safe level as described in **Section 4 - AMPLIFIER CAPACITOR BANK SAFETY.**

5-2 Removal of Fan (Continued)

2. Turn amplifier over and remove 18 screws from the lower panel of RF AMP and remove panel. See Illustration 5-1.

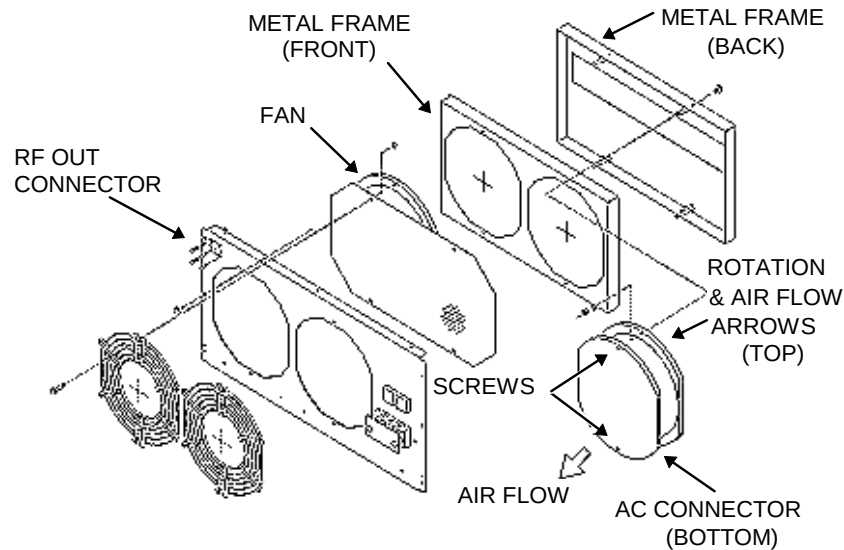


DISASSEMBLY OF FANS
ILLUSTRATION 5-1

3. Disconnect AC connectors from fans.
4. Using the pliers to hold nut, remove lower two screws from fan out take grills.
5. Remove four screws from the rear panel at the RF OUT (N-TYPE) connector.
6. Remove six screws on the rear panel (three screws along each side) and remove panel.
7. Using the pliers to hold nut, remove two screws from the top of the fan out take grills.
8. Remove screws from back side of metal frame to remove fans. Gently push fans out of metal frame (back).
9. Remove nuts securing fans to metal frame (front).

5-3 Installation of Fan

See Illustration 5-2.



FAN INSTALLATION
ILLUSTRATION 5-2

1. Place new fan on front metal frame with rotation arrow (pointing left) and air flow arrow (pointing to rear panel) positioned on top. The AC Connectors indicates the bottom of the fans.

Note

Be sure to install FAN allowing air to flow out from the back panel. The direction of the air flow is indicated by the arrow on the FAN and should point toward the rear panel.

2. Secure screws in top and bottoms of fans to front metal frame.
3. Insert metal panel with fans attached into the back metal frame.
4. Insert four screws and secure metal frame.
5. Put fans and frame into place and attach the rear panel and fan out take grills by inserting top two screws and securing to fans.
6. Insert and fasten four screws for the RF Out (N-TYPE) Connector. Make sure screws are inserted through rear panel, connector and metal plate before securing tightly.
7. Insert six screws and fasten rear panel.
8. Replace and fasten top RF Amplifier panel with (14) screws.
9. Turn amplifier over and insert screws into lower fan grill and attach fans to rear panel.
10. Attach the AC Connectors to bottom of fans.
11. Replace and fasten lower panel.

5-3 Installation of Fan (Continued)

- 12.** Perform the replacement procedure in **Section 3-2 Amplifier Replacement**.

6 - POWER SUPPLY

6-1 Tools Required

#1 Phillips Screwdriver

6-2 Removal of 24 Volt Power Supply

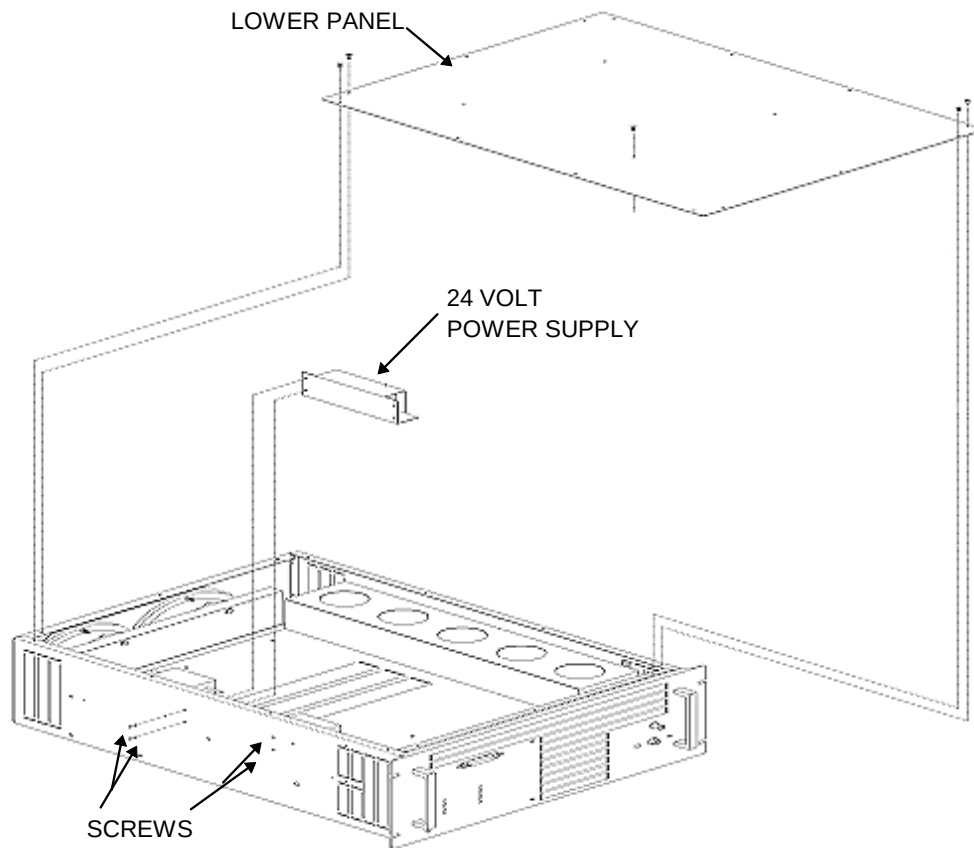


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1. Confirm that the 14 screws on the upper panel of the RF amplifier have been removed and that the capacitor bank has been discharged to a safe level as described in **Section 4 - AMPLIFIER CAPACITOR BANK SAFETY.**

6-2 Removal of 24 Volt Power Supply (Continued)

2. Replace the upper panel and screws after verifying that the capacitor bank is discharged.



24 VOLT POWER SUPPLY
ILLUSTRATION 6-1

3. Turn amplifier over and remove the lower panel. See Illustration 6-1.
4. Detach Power Supply from the right side panel by removing four screws.
5. Remove two connectors on each side of Power Supply.

6-3 Installation of 24 Volt Power Supply

1. Attach connectors to each side of Power Supply.
2. Attach Power Supply to side panel by inserting and fastening four screws.
3. Replace lower panel and fasten.
4. Perform the replacement procedure in **Section 3-2 Amplifier Replacement**.

7 – AMPLIFIER FUSE REPLACEMENT

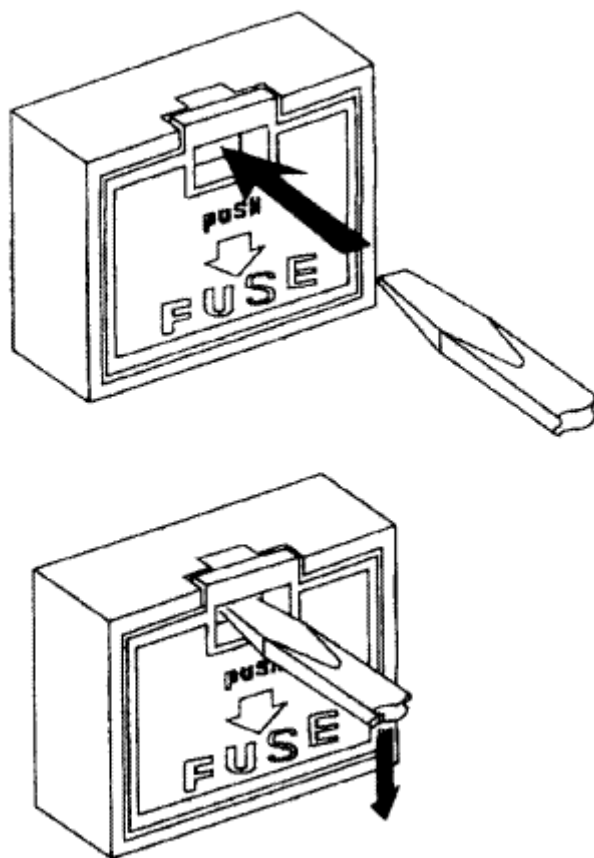
7-1 Tools Required

- Small slot screwdriver with blade no wider than 3mm.

7-2 Removal of Fuse

Two (2) plastic fuse housings are located on the upper, rear area of each RF amplifier.

1. From the rear, exterior of the amplifier insert the screw driver into the FUSE HOLDER. See Illustration 7-1.



REMOVAL OF FUSE HOLDER
ILLUSTRATION 7-1

2. Lift the driver slightly upward.
3. The Fuse Holder is released and pops out.
4. Pull fuse from the Fuse Holder.

7-3 Installation of Fuse and Fuse Holder

1. Push new fuse into end of Fuse Holder.
2. Push Fuse Holder into housing with finger, until it clicks.

APPENDIX A - UNIVERSAL LIFT KIT ASSEMBLY FOR 1.0T RF/PDU CABINETS

WARNING!

POSSIBLE PERSONAL INJURY. HEAVY OBJECT - THE HOIST KIT WEIGHS 73 LBS/33 KG. ASSISTANCE TO LIFT THIS PRODUCT [MECHANICAL (DOLLIES) OR ADDITIONAL PERSONNEL] IS REQUIRED.

A-1 Introduction

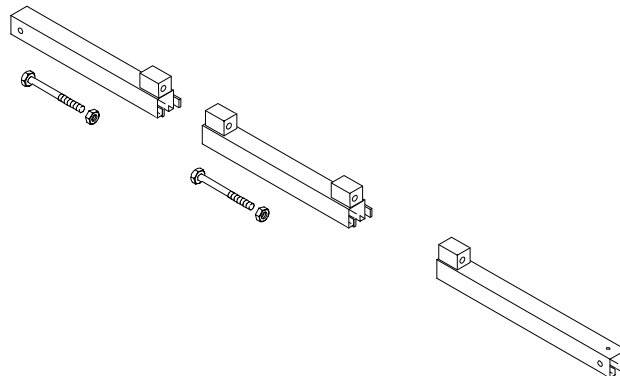
The Universal Lift Hoist is assembled from parts stored in the Universal Lift Hoist Kit. Lift attachments specially designed for the 1.0T RF/PDU Cabinet are found in the outer-most compartments of the lift kit. The lift brackets for the Teli amplifier are located in the upper right side of the lift kit and are part number 2187660.

A-2 Tools Required

- Universal Lift Hoist Kit with RF/PDU attachments, 46-317266G5
- Small hand tools including a Phillips-head screwdriver
- Two medium sized crescent wrenches

A-3 Assembly Procedural Steps

1. Locate the three (3) large Steel Rail sections and place them on a flat surface for assembly. See Illustration A-1.

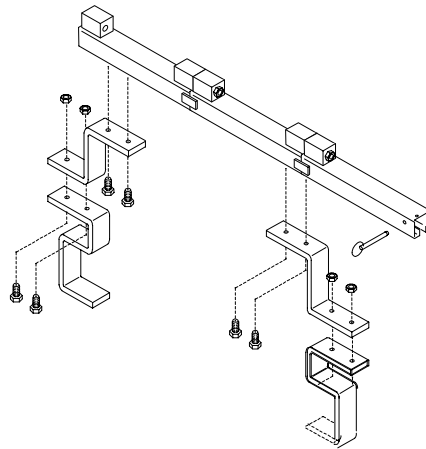


STEEL RAIL SECTIONS, HEX BOLTS, AND HEX NUTS
ILLUSTRATION A-1

2. Arrange the rail sections in the proper order (rear, center, then front) and fasten them together with 4.75" X 1/2" caps crews and hex nuts.
3. Use the crescent wrenches to tighten the cap screws.

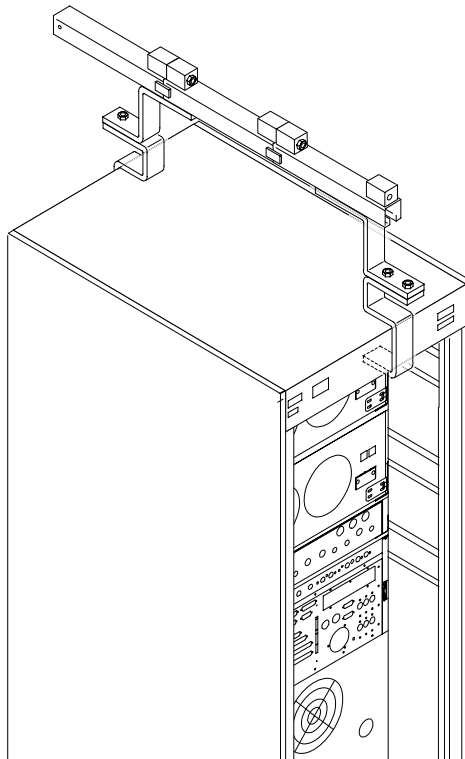
A-3 Assembly Procedural Steps (Continued)

4. Assemble and fasten the elevating brackets to the rail assembly. See Illustration A-2.



HOIST TOP RAIL PARTS WITH ELEVATING BRACKETS
ILLUSTRATION A-2

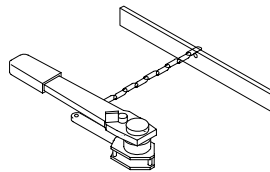
5. Open the front door and remove the rear cover of the SRF Cabinet
6. Place the Steel Rail and elevating bracket assembly at the top of the SRF Cabinet. Center the Rail Assembly on the top of the SRF Cabinet. See Illustration A-3.



HOISTPARTS REAR VIEW ASSEMBLED
ILLUSTRATION A-3

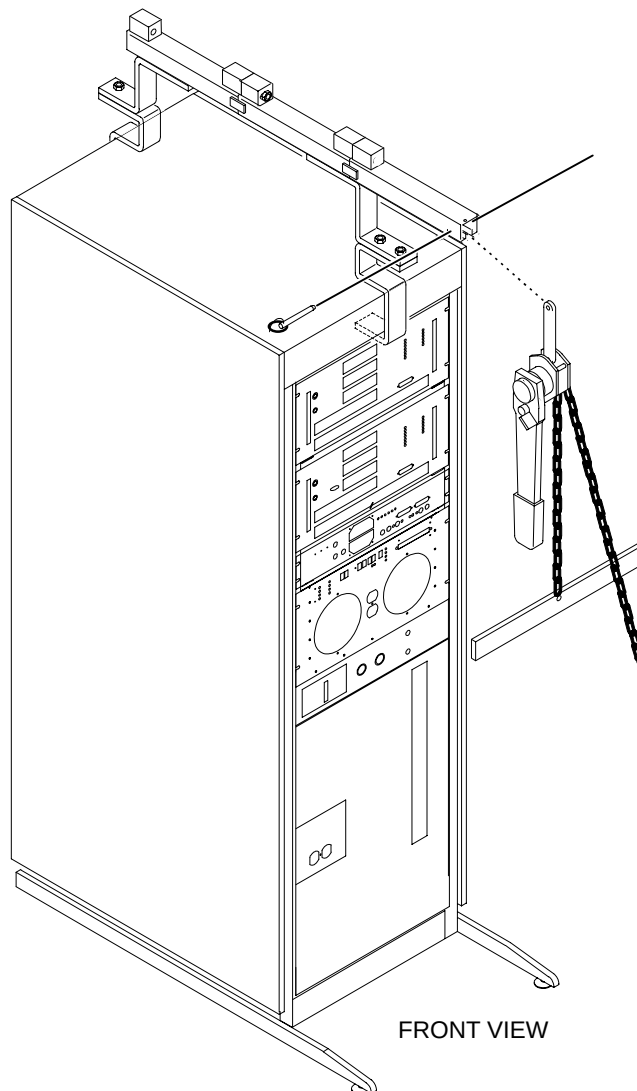
A-3 Assembly Procedural Steps (Continued)

7. Locate the Hoist Assembly in the Lift Kit. See Illustration A-4.



HOIST ASSEMBLY
ILLUSTRATION A-4

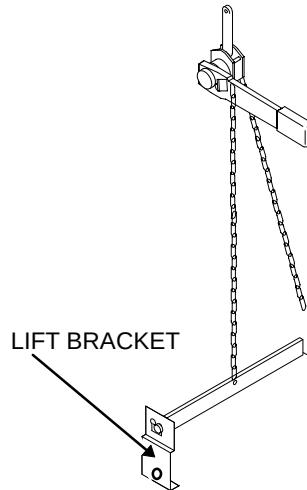
8. Insert the Hoist Assembly into the open end of the front rail. Insert a locking pin into the holes near the front edge of the front rail. This locking pin prevents the Hoist Assembly from sliding out of the rails. See Illustration A-5.



PLACEMENT OF HOIST ASSEMBLY INTO FRONT END OF STEEL RAIL ASSEMBLY
ILLUSTRATION A-5

A-3 Assembly Procedural Steps (Continued)

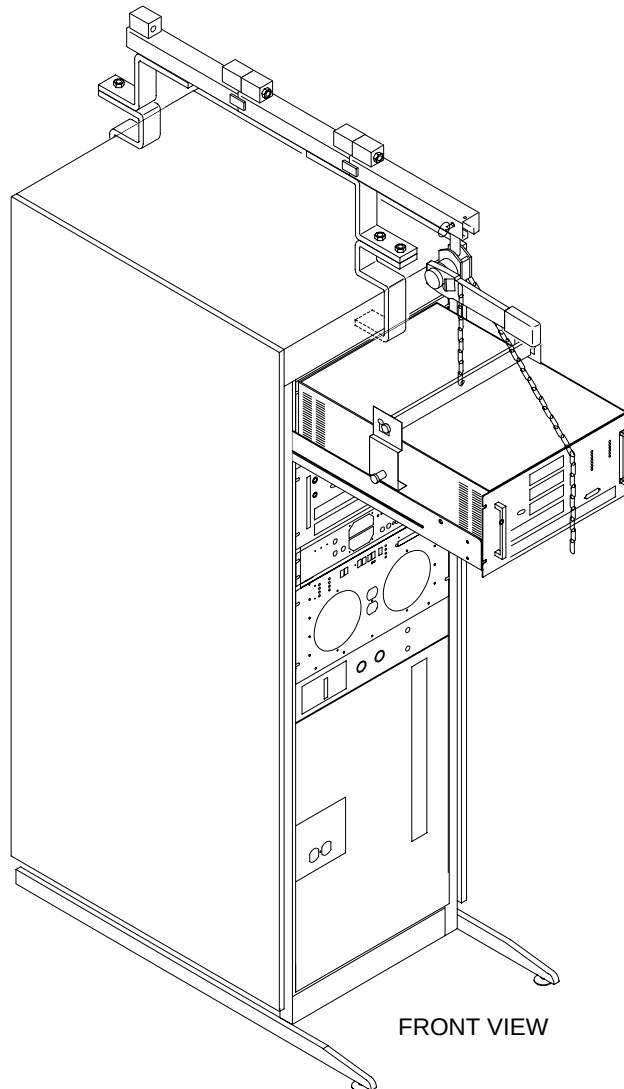
9. Select the proper lift brackets for your particular application and attach them to the Hoist Assembly. See Illustration A-6.



ATTACHING LIFT BRACKET TO HOIST ASSEMBLY
ILLUSTRATION A-6

A-3 Assembly Procedural Steps (Continued)

10. The Universal Lift Hoist is now ready for operation. Illustration A-7 shows the Universal Lift Hoist being used to remove a RF Amplifier Module from a RF/PDU Cabinet. Although the location of the RF components in the SRF Cabinet is different, the removal and installation process is the same as for the RF/PDU Cabinet.



USING THE UNIVERSAL LIFT HOIST TO REMOVE THE UPPER RF AMP
ILLUSTRATION A-7

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Mar 03, 1998 Apr. 06, 1998	K. Keshena	Preliminary version. Included 55 Volt Power Supply information.
B	May 12, 1998	wek	Updated lock out/tag out info
0	July 28, 1998	K. Keshena	Initial Release.
1	January 8, 2001	D. Thome	Removed procedures for FRU N parts. Added new Lockout/Tagout process.
2	May 4, 2001	D. Thome	Added information concerning hoist kit lift brackets.
3	May 15, 2001	D. Thome	Modified Lockout/Tagout information.
4	10/17/2007	K.Keshena	PSR 13105722, requires the warning statement for the Hoist Kit.